



VIT-AP
UNIVERSITY

Module 2: Cloud-Enabling Technologies

Data Center Technology

Mrs. T. Ramathulasi

ramathulasi.t@vitap.ac.in

OUTLINE

- **Cloud-Enabling Technologies**

Broadband Networks and Internet Architecture

foundational network and internet technologies that enable cloud computing, including the role of broadband in providing high-speed data access.

Data Center Technology

Covers the technology and infrastructure used in data centers, which are critical to cloud computing as they house the servers, storage, and networking equipment that deliver cloud services.

Virtualization Technology

that allows multiple virtual machines (VMs) to run on a single physical machine, enabling efficient resource utilization and scalability in cloud environments.

Web Technology

Focuses on the web-based technologies that support cloud computing, such as web servers, APIs, and protocols that facilitate interaction between cloud services and users.

Multitenant Technology

Describes the technology that allows multiple users or tenants to share the same computing resources in a cloud environment while keeping their data and applications isolated from each other.

Containerization

Discusses containerization, a lightweight form of virtualization where applications and their dependencies are packaged together into containers, enabling consistent deployment across different environments.

- **Summary**

- **Data Center Technology**

- ❑ Data centers are specialized facilities that house and manage IT resources efficiently.

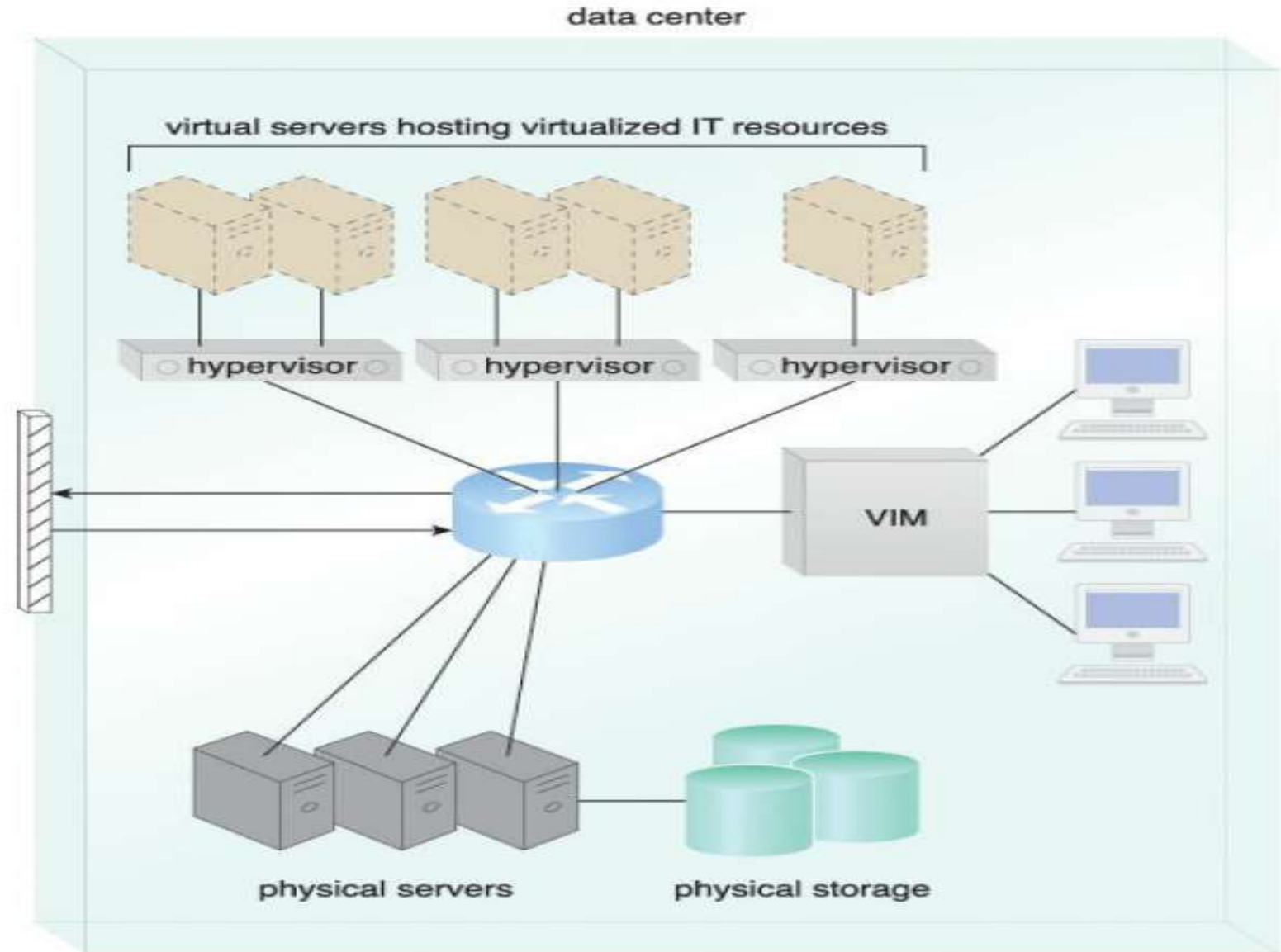
- ❑ Key components include:

- Servers: Physical or virtual machines that host applications and data.
 - Storage Systems: Devices like SAN, NAS, and DAS (Direct Attached Storage (DAS) Network Attached Storage (NAS) Storage Area Network (SAN)) used to securely store data.
 - Networking Equipment: Routers, switches, and firewalls that ensure connectivity and data flow.
 - Cooling Systems: CRAC (Computer Room Air Conditioner) units to manage temperature and prevent overheating.
 - Power Supply and Backup: UPS, generators, and PDUs to provide continuous power.
 - Security Systems: Physical and cyber protections, including access controls and encryption.
 - Virtualization Technologies: Software enabling multiple virtual instances on a single server.
 - DCIM (Data Center Infrastructure Management): Tools for monitoring and optimizing operations.
 - Telecommunication Systems: Infrastructure for internal and external data communication.
 - Cloud Integration: Integration with cloud technologies for scalability and flexibility.

- Virtualization
 - Data centers use both physical and virtual IT resources.
 - The physical layer includes infrastructure and hardware, while the virtualization layer abstracts these resources into virtual components, making them easier to manage and control.
- The common components of a data center working together to provide virtualized IT resources supported by physical IT resources.
- A hypervisor is a software that you can use to run multiple virtual machines on a single physical machine
- Understanding Virtual Infrastructure Management (VIM)

Data Center Technology

- ❑ Data Center Technology
- ❑ Virtualization



Data Center Technology

- ❑ Data Center Technology
- ❑ Virtualization
- ❑ Standardization and Modularity

- Standardization and Modularity

- Data centers use standardized hardware to support scalability and reduce costs.
- Modularity: Designed with modular architectures, allowing for easy expansion and quick hardware replacements.
- Consolidation: Fewer physical components are needed due to improved virtualization and device performance, enabling shared IT resources across systems and users.

Data Center Technology

- ❑ Data Center Technology
- ❑ Virtualization
- ❑ Standardization and Modularity
- ❑ Automation

- Automation

- Specialized platforms automate tasks such as provisioning, configuration, patching, and monitoring.
- Autonomic Computing: Enables self-configuration and self-recovery in data center management.
- Modern tools and technologies enhance automation, reducing the need for supervision.

Data Center Technology

- Remote Operation and Management

- Remote Management: IT resources are managed through networked remote consoles and systems.
- Minimal On-Site Visits: Technical staff only visit server rooms for specific tasks like equipment handling and maintenance.
- Efficiency: Reduces the need for physical presence, streamlining operations.

- ❑ Data Center Technology
- ❑ Virtualization
- ❑ Standardization and Modularity
- ❑ Automation
- ❑ Remote Operation and Management

Data Center Technology

- High Availability in Data Centers

- Redundant Power Supplies: Uninterruptible power supplies (UPS) and backup generators.
- Redundant Cabling: Multiple pathways for network and power to ensure connectivity.
- Environmental Controls: Duplicate cooling systems and humidity controls.
- Communication Links: Multiple network connections for continuous communication.
- Clustered Hardware: Load balancing with clustered servers for reliability.
- Failover Mechanisms: Systems designed to switch to backups in case of failure.

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- ❑ High Availability in Data Centers

Data Center Technology

- Security-Aware Design, Operation, and Management
 - Physical Access Controls: Secure entry points with access cards or biometric systems.
 - Logical Access Controls: Authentication and authorization measures for data and system access.
 - Data Recovery Strategies: Comprehensive backup and disaster recovery plans.
 - Outsourcing Considerations: Historically, outsourcing required long-term commitments and lacked flexibility.
 - Cloud Solutions: Address issues with features like ubiquitous access, on-demand provisioning, rapid elasticity, and pay-per-use models.

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- ❑ Security-Aware Design, Operation, and Management

- Facilities

- Custom-Designed Locations: Specialized areas for computing, storage, and network equipment.
- Functional Layout Areas: Specific zones for equipment, power supplies, and environmental controls.
- Power Supplies: Redundant and reliable power systems to ensure uninterrupted operation.
- Cabling: Structured and organized cabling for connectivity and efficiency.
- Environmental Control Stations: Manage heating, ventilation, air conditioning (HVAC), and fire protection.
- Segregated Spaces: Divided into distinct areas for different functions and equipment.

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Data Center Technology

- Computing Hardware in Data Centers

- Standardized Commodity Servers:

- High processing power and storage capacity.
 - Servers housed in standardized racks with interconnects for power, network, and cooling.
 - Hardware Architectures: Support for x86-32bit, x86-64, and RISC architectures.
 - Multi-Core CPUs: Power-efficient CPUs with hundreds of cores in compact rack units.
 - Redundant and Hot-Swappable Components: Includes hard disks, power supplies, network interfaces, and storage controllers.

- Blade Server Technologies:

- Blade Enclosures: Rack-embedded units for interconnections and shared resources.
 - Shared Power and Cooling: Optimizes space and power usage.
 - Hot-Swapping Capabilities: Allows for server replacement and maintenance without downtime.

- Management and Monitoring:

- Industry-Standard and Proprietary Software: Configures, monitors, and controls hardware from remote consoles.
 - Centralized Management Consoles: Enables oversight of large numbers of physical and virtual servers from a single interface.

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- ❑ Computing Hardware in Data Centers

Data Center Technology

- Storage Hardware

- Data centers use specialized storage systems to manage large volumes of digital information, typically involving:
 - Hard Disk Arrays: These divide and replicate data across multiple drives for improved performance and redundancy, often using RAID schemes.
 - I/O Caching: Enhances performance by caching data through hard disk array controllers.
 - Hot-Swappable Hard Disks: Allows disks to be replaced without powering down.
 - Storage Virtualization: Utilizes virtual disks and shared storage.
 - Fast Data Replication: Includes snapshotting and volume cloning for quick data recovery.
- Storage systems can be networked or directly attached (DAS). Networked storage includes:
 - Storage Area Network (SAN): Provides block-level storage access via a dedicated network using protocols like SCSI (Small Computer System Interface).
 - Network-Attached Storage (NAS): Manages hard drive arrays through a network using file-centric protocols such as NFS or SMB.

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- ❑ Computing Hardware in Data Centers
- ❑ Storage Hardware

Data Center Technology

- Network Hardware

- Data centers use extensive network hardware to support connectivity through five main subsystems:
 - Carrier and External Networks Interconnection: Includes backbone routers for WAN-LAN routing and security devices like firewalls and VPN gateways.
 - Web-Tier Load Balancing and Acceleration: Utilizes devices for web acceleration such as XML pre-processors, encryption appliances, and layer 7 switches for content-aware routing.
 - LAN Fabric: Provides internal high-performance connectivity with network switches that support speeds up to 10 Gbps and features like VLANs, link aggregation, and load balancing.
 - SAN Fabric: Connects servers and storage systems using technologies like Fibre Channel (FC), Fibre Channel over Ethernet (FCoE), and InfiniBand network switches.
 - NAS Gateways: Facilitates connectivity for NAS storage devices and performs protocol conversion between SAN and NAS systems.

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- ❑ Storage Hardware
- ❑ Network Hardware

Data Center Technology

- Other Considerations

- Technological Obsolescence:

- IT hardware needs replacement every 5-7 years, creating complexity in managing diverse equipment.
 - Virtualization can help manage this complexity.

- Security Concerns:

- Housing large volumes of data in a single data center increases the risk of a significant breach.

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- ☐ Network Hardware
- ☐ Other Considerations

Summary on Fundamental Concepts and Models : cloud deployment models.

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- ☐ Storage Hardware
- ☐ Network Hardware
- ☐ Other Considerations

References

1. Thomas Erl, Ricardo Puttini, Zaigham Mahmood, “Cloud Computing: Concepts, Technology & Architecture”, PHI Publications, 2013.

Quiz

Question: What is the primary purpose of a data center?

- A) To provide storage space for physical documents
- B) To house computer systems and associated components, such as telecommunications and storage systems
- C) To serve as a backup facility for manufacturing units
- D) To manage and distribute internet connectivity

Answer: A) Constant uptime with minimal service interruptions

Question: What is a key benefit of virtualization in data centers?

- A) Increased hardware dependency
- B) Better utilization of physical resources
- C) Increased energy consumption
- D) Reduction in data security

Answer: B) Better utilization of physical resource

Question: What is the benefit of standardization in data center components?

- A) It makes replacement parts more difficult to find.
- B) It simplifies maintenance and reduces costs.
- C) It increases the diversity of components used.
- D) It requires more customization for each data center.

Answer: B) It simplifies maintenance and reduces costs.

**Thank
You**